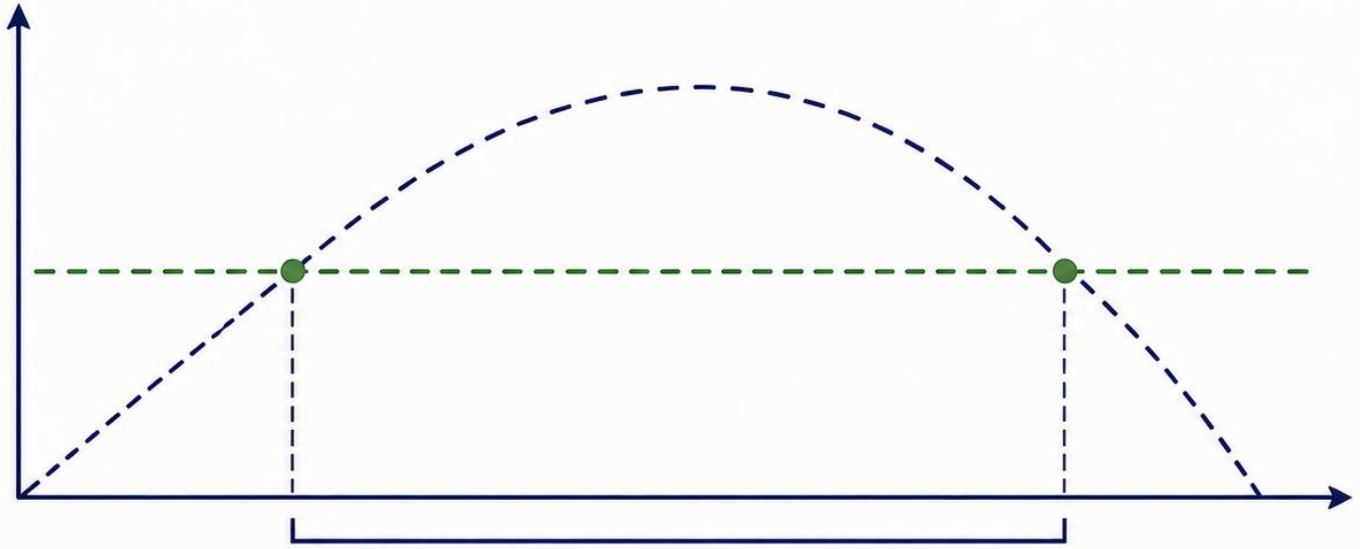
PHY4605 · WEEK 05 · LECTURE

Root Finding

Locate a physical solution and verify the residual



Model → Units → Algorithm → Code → Validation → Interpretation

Turn the target condition into a residual

PHYSICAL MODEL

$$R(\theta) = \frac{v_0^2}{g}\sin(2\theta), \quad v_0 = 20\,\mathrm{m\,s^{-1}}, \quad g = 9.81\,\mathrm{m\,s^{-2}}$$

$$R(\theta) = 35\ \mathrm{m}$$

TARGET CONDITION

$$R(\theta) = 35\ \mathrm{m}$$

RESIDUAL ZERO

$$f(\theta) = R(\theta) - 35\ \mathrm{m} = 0$$



Level-ground projectile range model gives the range $R(\theta)$ for launch angle θ.



We want the range to hit the target value of 35 m.



Define residual as model minus target. A root makes the residual zero.



- **Original model:** $R(\theta) = \frac{v_0^2}{g}\sin(2\theta),$
- **Target condition:** $R(\theta) = 35\ \mathrm{m}.$
- **Residual:** $f(\theta) = R(\theta) - 35\ \mathrm{m}.$
- **A root satisfies** $f(\theta) = 0.$



Variables and units

θ = launch angle (degree)
$R(\theta)$ = range (m)
$f(\theta)$ = residual (m)

Root unit
degree

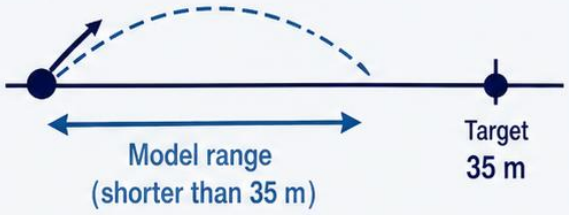
Residual unit
m

Read the residual sign as physics

The residual compares the model's range to the 35 m target: $f(\theta) = \text{range}(\theta) - 35\ \text{m}$.

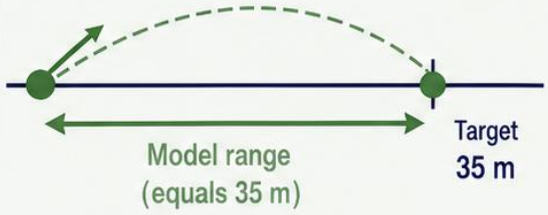
Undershoot ($f(\theta) < 0$)

Model range is **shorter** than the target.



Root ($f(\theta) = 0$)

Model range **equals** the target.



Overshoot ($f(\theta) > 0$)

Model range is **longer** than the target.



$$
\begin{aligned}
f(\theta) < 0 &\implies \text{model range is shorter: undershoot} \\
f(\theta) = 0 &\implies \text{model range equals target: root} \\
f(\theta) > 0 &\implies \text{model range is longer: overshoot}
\end{aligned}
$$

Residual unit: metres (m)

Quick check

Which sign means the model's range is longer than the target?

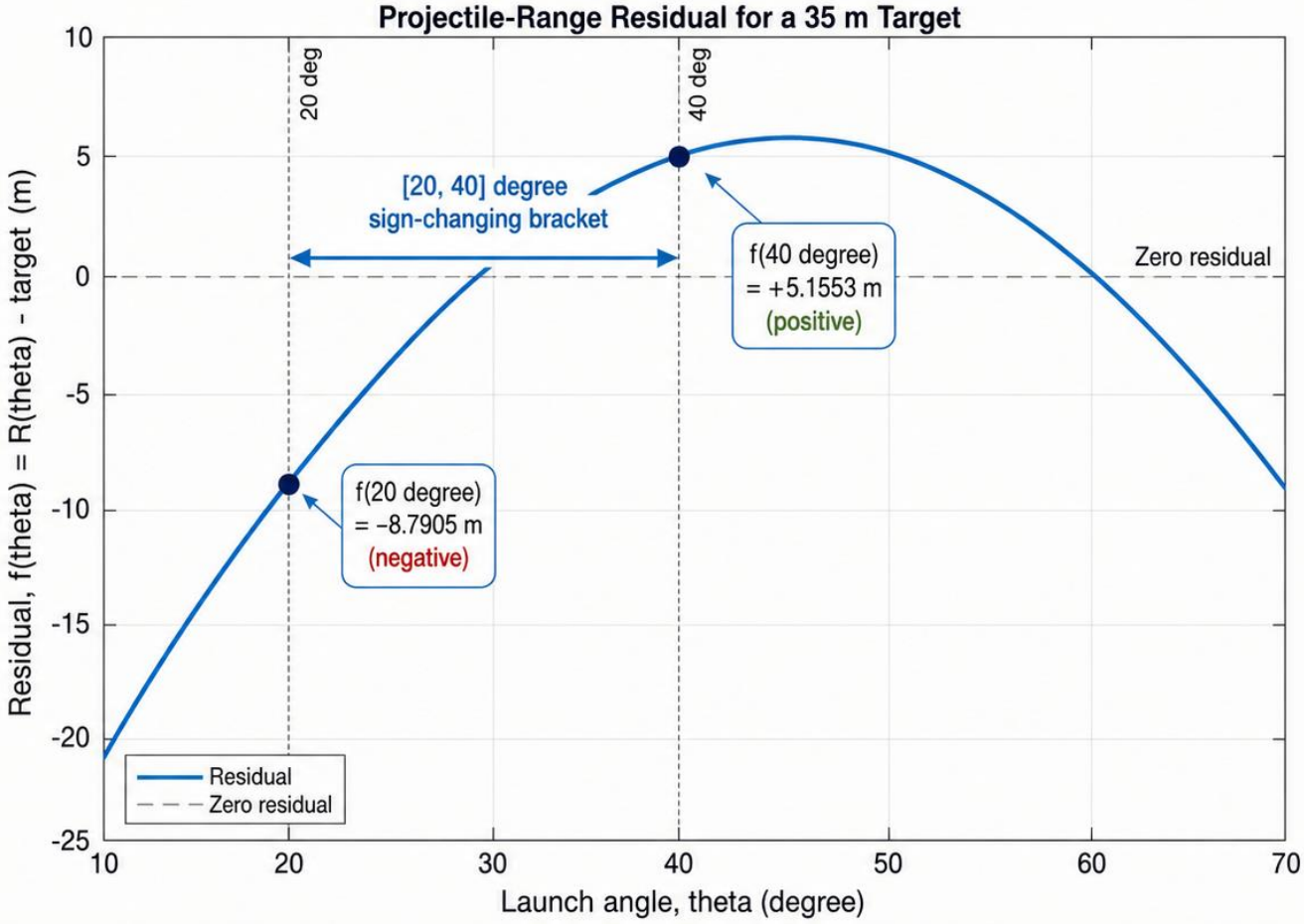
Find the zero crossing before coding

Bracket evidence (low angle)

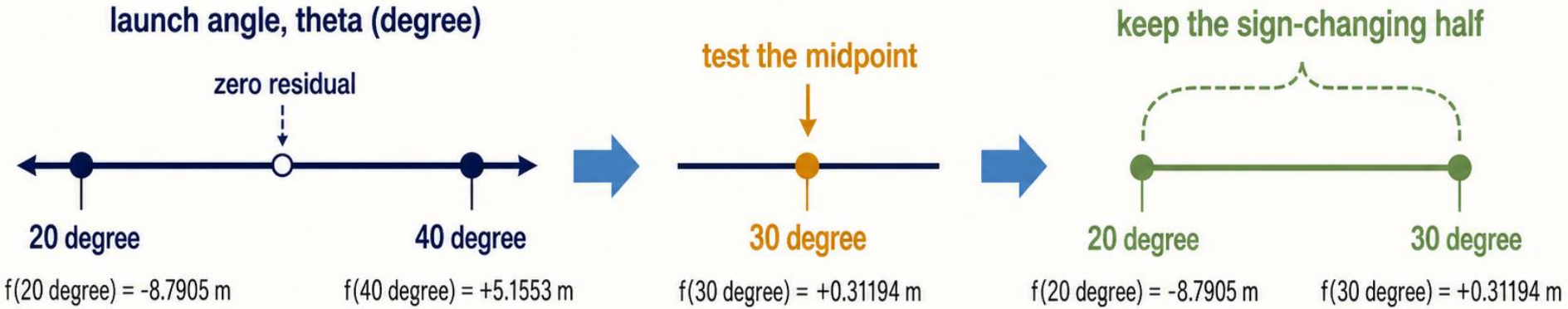
- Scan theta = 10:1:70 degree.
- f(20 degree) = −8.7905 m approximately.
- f(40 degree) = +5.1553 m approximately.
- The sign-changing bracket [20, 40] degree contains the low-angle zero crossing.

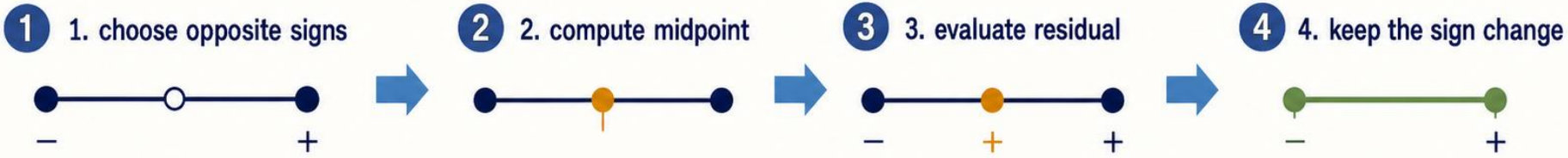
Left endpoint (undershoot)	Right endpoint (overshoot)
20 degree	40 degree
f(20 degree)	f(40 degree)
−8.7905 m	+5.1553 m
negative	positive

Where does f(theta) cross zero?
Look between 20 and 40 degree.



Bisection keeps a sign change alive



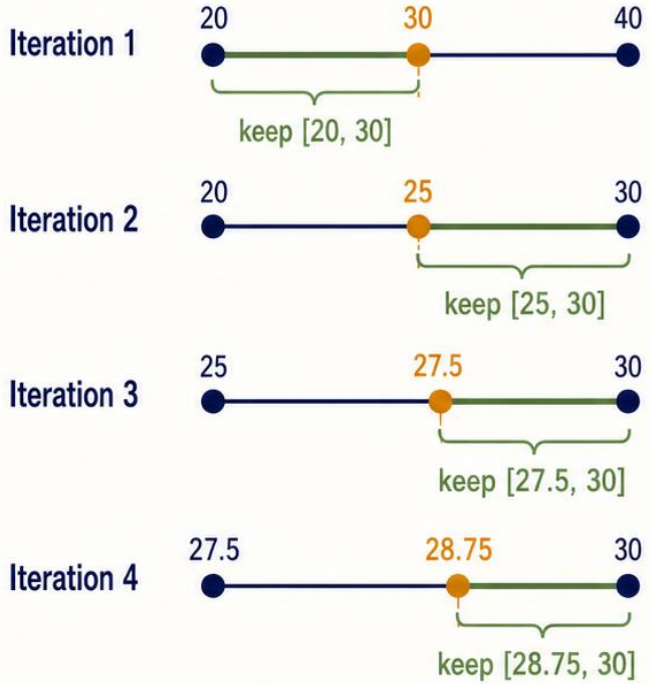


Why is the midpoint not enough?

Because bisection needs a bracket whose endpoint residuals have opposite signs.

Trace the first four bisection updates

iteration	current bracket (degree)	midpoint (degree)	f(mid) (m)	keep
1	[20, 40]	30	+0.31194	[20, 30]
2	[20, 30]	25	-3.7648	[25, 30]
3	[25, 30]	27.5	-1.5993	[27.5, 30]
4	[27.5, 30]	28.75	-0.61095	[28.75, 30]



1. 1. Iteration 1 keeps [20, 30] after testing 30 degree.
2. 2. Iteration 2 keeps [25, 30] after testing 25 degree.
3. 3. Iteration 3 keeps [27.5, 30] after testing 27.5 degree.
4. 4. Iteration 4 keeps [28.75, 30] after testing 28.75 degree.

? Which half still contains opposite endpoint signs?

Read the bisection scaffold, line by line

```
bis_left_deg = left_deg;
bis_right_deg = right_deg;
for iteration = 1:max_iterations
    mid_deg = (bis_left_deg + bis_right_deg)/2;
    f_mid_m = (v0_mps^2/g_mps2)*sind(2*mid_deg) - target_m;
    if abs(f_mid_m) < tolerance_m
        break
    end
    f_bis_left_m = (v0_mps^2/g_mps2)*sind(2*bis_left_deg) - target_m;
    if f_bis_left_m*f_mid_m < 0
        bis_right_deg = mid_deg;
    else
        bis_left_deg = mid_deg;
    end
end
bisection_root_deg = mid_deg;
```

MIDPOINT —
mid_deg creates the trial angle

RESIDUAL —
f_mid_m is measured in metres

STOP —
abs(f_mid_m) < tolerance_m

KEEP SIGN CHANGE —
the sign-product test moves one endpoint



Which line protects the sign-changing bracket?

Stop once, then substitute back

Bisection result (low-angle root)

- Bisection gives theta = 29.5679 degree after 13 recorded iterations.
- The final residual is +4.1164e-4 m.
- Substitution gives a model range of 35.0004 m.

Use abs(residual_m) < 1e-3 m as the one Core stopping rule.

Substitution check (validation)

$$\theta_* = 29.5679\,\text{degree}$$

$$R(\theta_*) = 35.0004\,\text{m}$$

$$f(\theta_*) = R(\theta_*) - 35\,\text{m} = +4.1164 \times 10^{-4}\,\text{m}$$

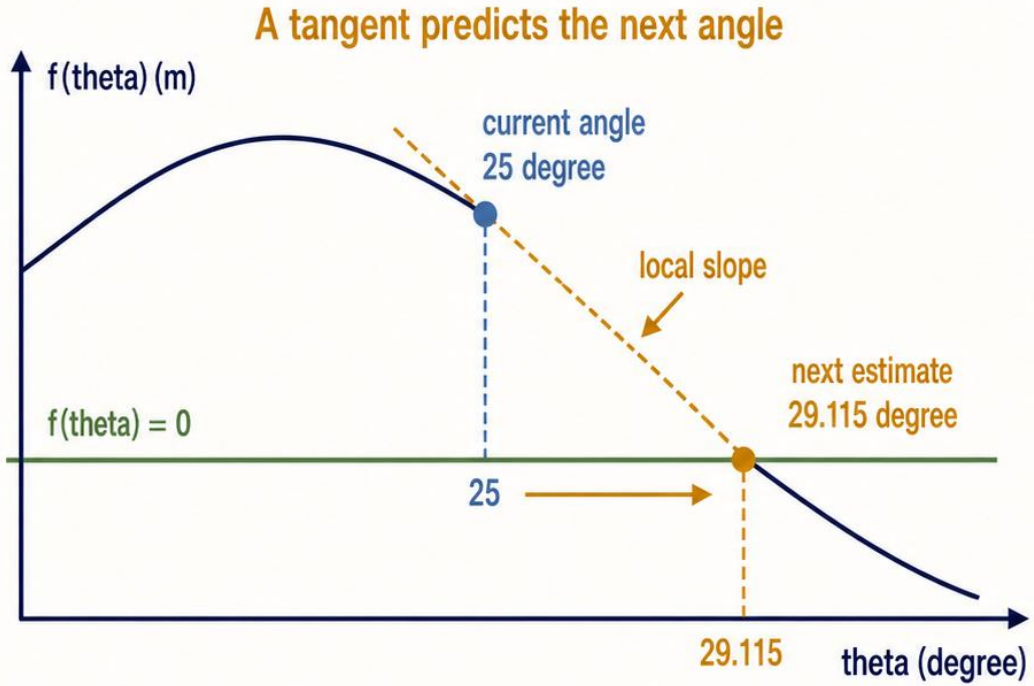
check	result	interpretation
root angle	29.5679 degree	low-angle launch angle
model range	35.0004 m	target condition is met
residual	$+4.1164 \times 10^{-4}$ m	below 10^{-3} m tolerance

Physical interpretation

At $\theta = 29.5679°$, the projectile's predicted range is 35.0004 m, which is within 10^{-3} m of the target.

Newton follows the local tangent

One estimate plus a local slope gives the next angle



$$\text{new angle} = \text{current angle} - \frac{\text{current residual}}{\text{current residual slope}}$$

$$\frac{df}{d\theta} = \frac{v_0^2}{g}\cos(2\theta)\,\frac{2\pi}{180} \qquad \text{(metres per degree)}$$

m/° slope unit = metres per degree

? What does the slope measure?

Trace Newton from 25 degree

MATLAB Newton scaffold (exact code)

```
newton_theta_deg = newton_start_deg;
newton_history = zeros(max_iterations,4);
for iteration = 1:max_iterations
    f_current_m = (v0_mps^2/g_mps2)*sind(2*newton_theta_deg) - target_m;
    slope_m_per_deg = (v0_mps^2/g_mps2)*cosd(2*newton_theta_deg)*(2*pi/180);
    newton_history(iteration,:) = [iteration, newton_theta_deg, f_current_m, slope_m_per_deg];
    if abs(f_current_m) < tolerance_m
        break
    end
    newton_theta_deg = newton_theta_deg - f_current_m/slope_m_per_deg;
end
newton_root_deg = newton_theta_deg;
```

- Start from theta = 25 degree.
- Each evaluation records angle, residual, and slope.
- The update moves from 25 to 29.115 to 29.562 to 29.567 degree.
- The fourth recorded residual is −7.1308e−7 m, so the one stopping rule is met.

Newton iteration trace (exact values)

evaluation	θ (degree)	residual (m)	slope (m/degree)	next angle (degree)
1	25	-3.7648	0.91488	29.115
2	29.115	-0.33461	0.74939	29.562
3	29.562	−0.0042242	0.73044	29.567
4	29.567	-7.1308×10^{-7}	0.73019	stop

Stop: abs(residual_m) < 1e−3 m is satisfied.
Final recorded residual = −7.1308e−7 m.

Working exposure: same root, different route

- Bisection reaches the locked tolerance in 13 recorded iterations.
- Newton reaches the same low-angle root in 4 recorded evaluations from the chosen start.
- Bisection preserves a sign-changing bracket.
- Newton needs a useful slope and a reasonable start.
- Both still require residual validation.



	BRACKETED	LOCAL
method	Bisection	Newton
update idea	midpoint and sign-changing half	current − residual/slope
recorded evaluations	13	4
what it preserves / needs	preserves a bracket	needs a useful slope and start



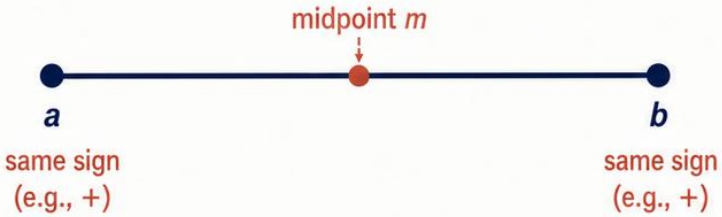
Both still require residual validation.

Substitute the reported angle back into the range model before interpreting the result.

A midpoint is not enough: the bracket must change sign

SAME-SIGN bracket (does not start bisection)

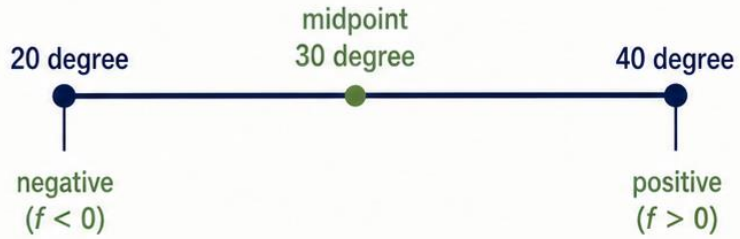
Both endpoint residuals have the same sign.
This interval does not provide Core bracket evidence.



Blocked start
Midpoint arithmetic is runnable,
but there is no Core bracket evidence.

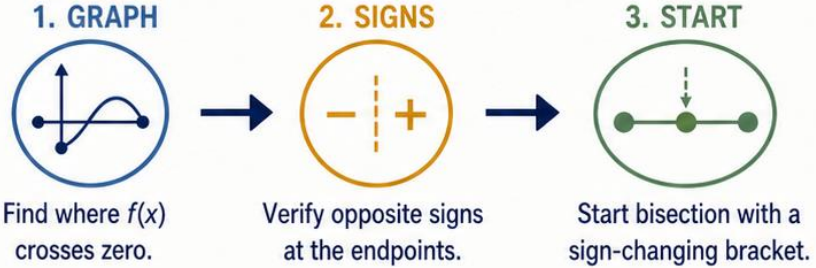
SIGN-CHANGING bracket (valid start)

Endpoint residuals have opposite signs.
This interval provides Core bracket evidence.



Valid start
Opposite signs guarantee a zero crossing
in the interval (Intermediate Value Theorem).

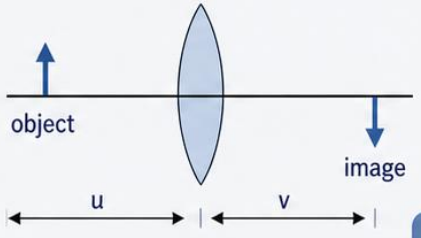
- Bisection starts with endpoint residuals of opposite signs.
- A same-sign interval can still produce a midpoint,
  but it does not give the Core bracket evidence.
- Check the graph and endpoint signs before iterating.
- Repair path: GRAPH -> SIGNS -> START.



Transfer the residual pattern to new physics

- Thin lens: residual $1/f - 1/u - 1/v$; root v in m; residual in m^{-1}.
- RC charging: residual $V_s(1-\exp(-t/(RC))) - V_{\text{threshold}}$; root t in s; residual in V.
- The shared pattern is **model -> residual -> bracket/start -> update -> stop -> substitute -> interpret.**

THIN LENS



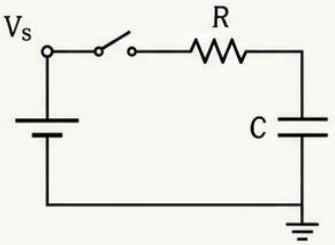
thin lens:

$$f(v) = \frac{1}{f} - \frac{1}{u} - \frac{1}{v} = 0$$

$[f(v) \text{ in } \mathrm{m}^{-1},\ v \text{ in m}]$

root: v in m | residual in m^{-1}

RC CHARGING



RC charging:

$$f(t) = V_s\left(1 - e^{-t/(RC)}\right) - V_{\text{threshold}} = 0$$

$[f(t) \text{ in V},\ t \text{ in s}]$

root: t in s | residual in V

SHARED ROOT-FINDING WORKFLOW

1. **model**
2. **residual** $f(x)$
3. **bracket/start**
4. **update**
5. **stop** $|f(x)| < 1\text{e-}3$ (residual rule)
6. **substitute** (validate)
7. **interpret** (unit & meaning)

inverse-length unit: m^{-1} | voltage unit: V | time unit: s

Takeaway and Next Step



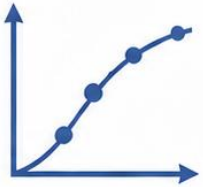
CORE IDEA

A root is an input whose residual reaches zero.



CHECK

Substitute the candidate input and inspect the residual.



NEXT STEP

Estimate a physical rate from nearby values.



What changed?



How do you know?